DO NOW — RECALL

Lesson 141 · ACID Transactions, Locking and Redundanc

Name: _____ Class: _____ Date: _____

1. What does foreign-key referential integrity prevent?

2. Which is faster in the worst case on sorted arrays, linear or binary search?

3. Which logic gate outputs 1 only when both inputs are 1?

4. What is the accumulator for?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A child key pointing to a nonexistent parent.

2. Model answer

Binary search: O(log n) rather than O(n).

3. Model answer

AND.

4. Model answer

Holding a CPU's working data/ALU results.

By the end, you can…

• Move £20: subtract account A and add account B.

• Both updates commit together or both roll back.

• Isolation/locks reduce interference; durable commits survive failures.

Vocabulary — know these terms

transaction — Group of DB operations treated as one unit of work.

ACID — Atomicity, Consistency, Isolation, Durability.

record locking — Restricting conflicting access while data is being changed.

redundancy — Duplicated resources/data for resilience, distinct from backup.

NEW LEARNING

Read, annotate and explain

Begin transaction

Move £20: subtract account A and add account B.

Protect consistency

Both updates commit together or both roll back.

Coordinate concurrent work

Isolation/locks reduce interference; durable commits survive failures.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A transfer subtracts £20 from Account A but crashes before adding £20 to Account B. Which ACID property matters most?

Subtract £20 from A

Identify the relevant information and make a prediction.

Crash before adding to B

Explain the change using correct technical terms.

Rollback both changes

Atomicity means the transfer is all-or-nothing. A DBMS transaction can roll back partial changes. Consistency checks valid constraints; isolation keeps concurrent transfers safe; durability preserves committed transfers.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Expand ACID and explain atomicity and durability. [6]

Question 2

Explain why two concurrent users editing one record can cause a lost update. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate replication, RAID and backups for database resilience. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

RAID or replication guarantees recovery after an accidental DELETE.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.